

ACHIEVE ACADEMY MATH HUB EXAM SOLUTIONS

ZIMSEC O'LEVEL

MATHEMATICS

4004/2

SPECIMEN P2 MARKING GUIDE

SYLLABUS B

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REMEMBER:

Decimal answers which are not exact should be given correct to three significant figures unless stated otherwise

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INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each answer.

This Marking Guide consists of 15 printed pages

Achieve Academy Math Hub

SECTION A

1. (a) LCM of 24 and 63

2	24
2	12
2	6
3	3
	1

3	63
3	21
7	7
	1

$24 = 2^3, 3$

$63 = 3^2, 7$

$LCM = 2^3 \times 3^2 \times 7$

$= 8 \times 9 \times 7$

$= 8 \times 63$

$= \mathbf{504} \quad [3]$

- (b) John 7 Peter 5

$$= \left(\frac{7}{12} \times 60\right) - \left(\frac{5}{12} \times 60\right)$$

$$= \$35 - \$25$$

$$= \mathbf{\$10} \quad [3]$$

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- (c)
- $2(a + 3b) - 5(4a - b)$

$$= 2a + 6b - 20a + 5b$$

$$= \mathbf{11b - 18a} \quad [3]$$

2. (a) Circumference

$$C = 2\pi r$$

$$= 2 \times \frac{22}{7} \times \frac{3.5}{2}$$

$$= \mathbf{11cm} \quad [2]$$

- (b) (i) HCF of
- $12a^3b^2$
- and
- $15a^4b^3$

$$12a^3b^2 = 2^2 \times 3 \times a^3 \times b^2$$

$$15a^4b^3 = 3 \times 5 \times a^4 \times b^3$$

$$HCF = 3 \times a^3 \times b^2$$

$$= 3a^3b^2 \quad [2]$$

(ii) LCM of $12a^3b^2$ and $15a^4b^3$

$$12a^3b^2 = 2^2 \times 3 \times a^3 \times b^2$$

$$15a^4b^3 = 3 \times 5 \times a^4 \times b^3$$

$$LCM = 2^2 \times 3 \times 5 \times a^4 \times b^3$$

$$= 60a^4b^3 \quad [2]$$

C. (i) $\frac{a-b}{c}$

$$= \frac{3-(-5)}{4}$$

$$= \frac{8}{4}$$

$$= 2 \quad [2]$$

(ii) $3x - 7 = 5x + 9$

$$-7 - 9 = 5x - 3x$$

$$-\frac{16}{2} = \frac{2x}{2}$$

$$x = -8 \quad [2]$$

3. (a) Total cost

$$= 2064 + 24 \times 164$$

$$= \$6\,000 \quad [3]$$

(b) $\frac{2}{7}$

$$= \frac{2^3}{7^3}$$

$$= \frac{8}{343} \times 1\,029$$

$$= 24cm^3 \quad [3]$$

(c) (i) $x + 15 + x = 45$

$$2x = 45 - 15$$

$$\frac{2x}{2} = \frac{30}{2}$$

$$x = 15 \quad [2]$$

(ii) $(X \cup Y)' \quad [2]$

4. (a) \$1 ZWD = R1.44

$$\text{\$1 000ZWD} = x$$

$$x = 1.44 \times 1\,000$$

$$= \mathbf{R1\,440} \quad [2]$$

(b) $\frac{1}{3}x = y \quad (1)$

$$2x - 3y = 1 \quad (2)$$

Using Substitution method

Make x subject of formula on equation (1)

$$\frac{1}{3}x = y \quad (3)$$

$$x = 3y$$

Now substitute equation (3) to equation (2)

$$2x - 3y = 1$$

$$2(3y) - 3y = 1$$

$$6y - 3y = 1$$

$$\frac{3y}{3} = \frac{1}{3}$$

$$y = \frac{1}{3}$$

Substitute y value on equation (3)

$$x = 3y$$

$$x = 3\left(\frac{1}{3}\right)$$

$$x = 1$$

$$\therefore x = 1 \text{ \& } y = \frac{1}{3} \quad [3]$$

(c) (i) 3Fig Bearing = 045° [1]

(ii) $\tan \theta = \frac{\text{Opp}}{\text{Adj}}$

$$\tan 60 = \frac{12}{x}$$

$$x = \frac{12}{\tan 60}$$

$$x = \mathbf{6.93\,cm} \quad [2]$$

(d) Number of sides = $\frac{360}{\text{Exterior Angle}}$

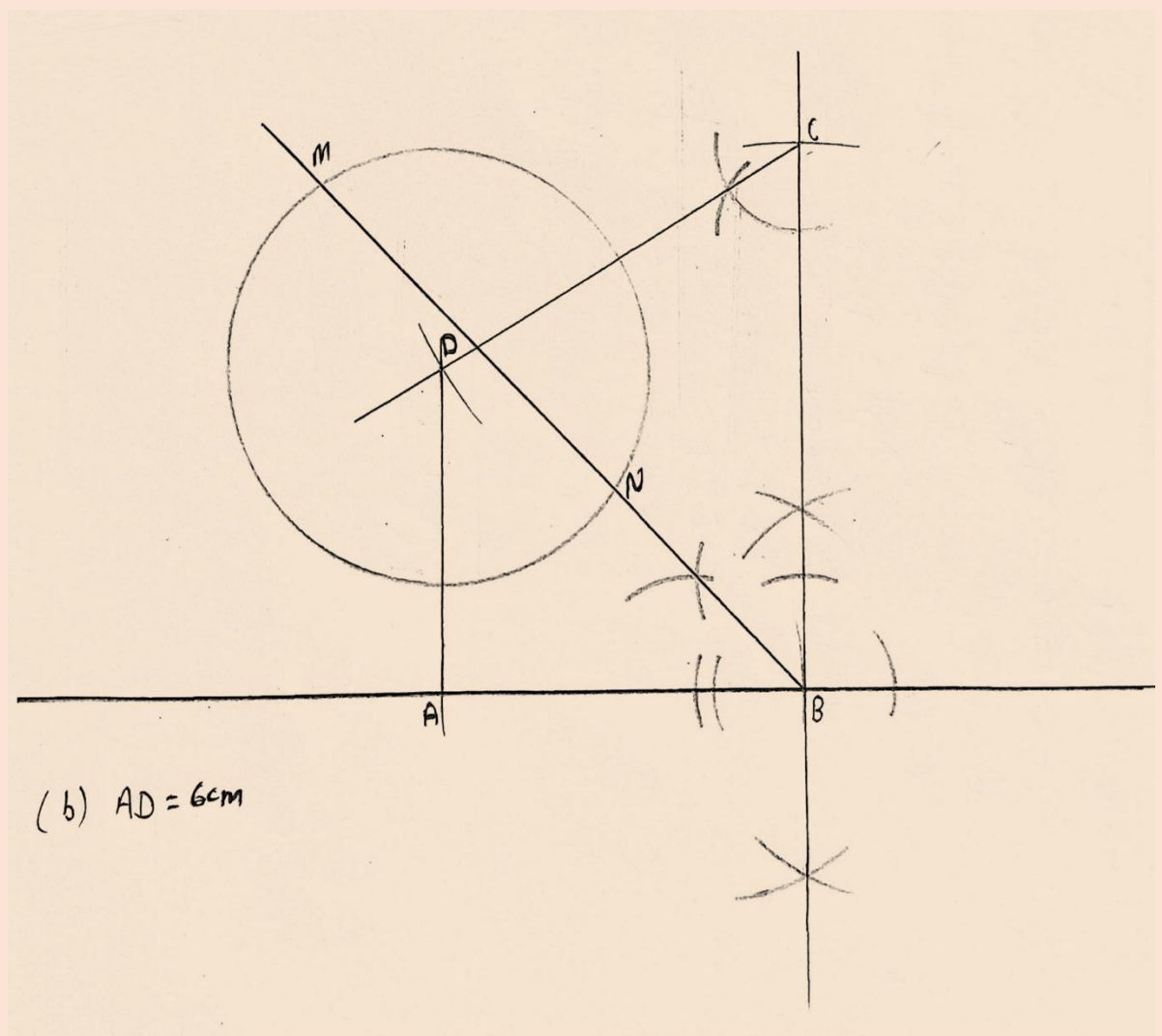
$$= \frac{360}{8}$$

$$= \mathbf{45\,sides} \quad [3]$$

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5.



(b) $AD = 6\text{cm}$

SECTION B

6. (a) LCM of
- $y^2 - 1$
- &
- $y^2 - 3y - 4$

$$y^2 - 1 = (y + 1)(y - 1)$$

$$y^2 - 3y - 4 = (y - 4)(y + 1)$$

$$LCM = (y + 1)(y - 1)(y - 4) \quad [3]$$

- (b) (i)
- $AOB \cong DOC$
- [1]

$$(ii) Area = \frac{5^2}{2^2} \times 12$$

$$= 75cm^2 \quad [2]$$

- (c) (i) Modal Mass = 7 [1]

$$(ii) \text{Mean Mass} = \frac{(5 \times 4) + (6 \times 3) + (7 \times 6) + (8 \times 2)}{15}$$

$$= 6\frac{2}{5} \quad [2]$$

- (d) (i) Chipu fails =
- $1 - \frac{2}{3}$

$$= \frac{1}{3} \quad [1]$$

- (ii) Both fails =
- $\frac{1}{4} \times \frac{1}{3}$

$$= \frac{1}{12} \quad [2]$$

7. (a) (i)
- $\frac{1}{2} + \frac{1}{y+1}$

$$= \frac{y+1+2}{2(y+1)}$$

$$= \frac{y+3}{2(y+1)} \quad [2]$$

- (ii) Hence solve

$$\frac{y+3}{2(y+1)} = 0$$

$$y + 3 = 0$$

$$y = -3 \quad [2]$$

- (b) (i)
- $Area = \frac{1}{2}bh$

$$16 = \frac{1}{2}(2x)(6x - 10)$$

$$16 = x(6x - 10)$$

$$16 = 6x^2 - 10x$$

$$\frac{6x^2}{2} - \frac{10x}{2} - \frac{16}{2} = 0$$

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$$3x^2 - 5x - 8 = 0 \text{ Shown } [2]$$

(ii) Solve

$$3x^2 - 5x - 8 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{5 \pm \sqrt{(-5)^2 - 4(3)(-8)}}{2(3)}$$

$$x = \frac{5 \pm \sqrt{121}}{6}$$

$$x = \frac{5+11}{6} \text{ or } \frac{5-11}{6}$$

$$x = \frac{8}{3} \text{ or } -1$$

$$x = -1 \text{ or } 2.67 [5]$$

(iii) Hence perpendicular Distance = $(6(2.67) - 10)$

$$= 16.02 - 10$$

$$= 6.02 \text{ cm } [1]$$

8. (a) (i) Relationship between P and Q

$$P \propto \frac{1}{Q}$$

$$P = \frac{k}{Q}$$

$$24 = \frac{k}{30}$$

$$k = 720$$

$$P = \frac{720}{Q} [3]$$

(ii) Value of Q

$$P = \frac{720}{Q}$$

$$3 = \frac{720}{Q}$$

$$Q = \frac{720}{3}$$

$$Q = 240 \text{ minutes } [1]$$

(b) (i) Value of V

$$Acc = \frac{\text{Change in Velocity}}{\text{change in Time}}$$

$$Acc = \frac{V}{5}$$

$$3 = \frac{V}{5}$$

$$V = 15m/s \quad [2]$$

(ii) Distance cover during first 12 seconds

Distance = Area under the graph

= Area of Trapezium

$$= \frac{1}{2}(a + b)h$$

$$= \frac{1}{2}(7 + 12)15$$

$$= 142\frac{1}{2}m \quad [2]$$

(iii) Value of T

$$165 = \frac{1}{2}(7 + T)15$$

$$\frac{330}{15} = 7 + T$$

$$22 = 7 + T$$

$$T + 22 = 7$$

$$T = 15seconds \quad [2]$$

(c) Total surface area of a cylinder

$$A = 2\pi r(r + h)$$

$$= 2 \times 3.142 \times 10(10 + 16)$$

$$= 2 \times 3.142 \times 10 \times 26$$

$$= 1633.84cm^2 \quad [2]$$

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9. (a) $S = c + kN$

$$60 = c + 25k \quad (1)$$

$$40 = c + 15k \quad (2)$$

Using substitution method

Make c subject on equation (1)

$$c = 60 - 25k \quad (3)$$

Substitute equation (3) on equation (2)

$$40 = c + 15k$$

$$40 = 60 - 25k + 15k$$

$$40 - 60 = -10k$$

$$\frac{-20}{-10} = \frac{-10k}{-10}$$

$$\underline{k = 2}$$

Substitute k value on equation ③

$$c = 60 - 25k$$

$$c = 60 - 25(2)$$

$$\underline{c = 10}$$

$$S = c + kN$$

$$S = 10 + 2(10)$$

$$= \$30 \quad [4]$$

(b) (i) $BD^2 + DC^2 = BC^2$

$$6^2 + DC^2 = 13^2$$

$$DC^2 = 169 - 36$$

$$\underline{DC = 11.53cm} \quad [2]$$

(ii) $\cos \theta = \frac{6}{13}$

$$\theta = \cos^{-1} \frac{6}{13}$$

$$\theta = 62.51^\circ$$

$$\text{total angle} = 33 + 62.51$$

$$= 95.51^\circ \quad [4]$$

(c) $A + B$

$$= \begin{pmatrix} -7 + 5 & -1 + (-2) \\ -3 + 1 & 4 + (-6) \end{pmatrix}$$

$$= \begin{pmatrix} -2 & -3 \\ -2 & -2 \end{pmatrix} \quad [2]$$

10. (a) (i) Value of p

$$p = 100 - (5 + 30 + 25 + 12 + 5) \\ = \mathbf{23} \quad [1]$$

$$\begin{aligned} \text{(ii) Mean time} &= \frac{((5 \times 5) + (15 \times 30) + (25 \times 23) + (35 \times 25) + (45 \times 12) + (55 \times 5))}{100} \\ &= \frac{2740}{100} \\ &= \mathbf{27.4 \text{ minutes}} \quad [2] \end{aligned}$$

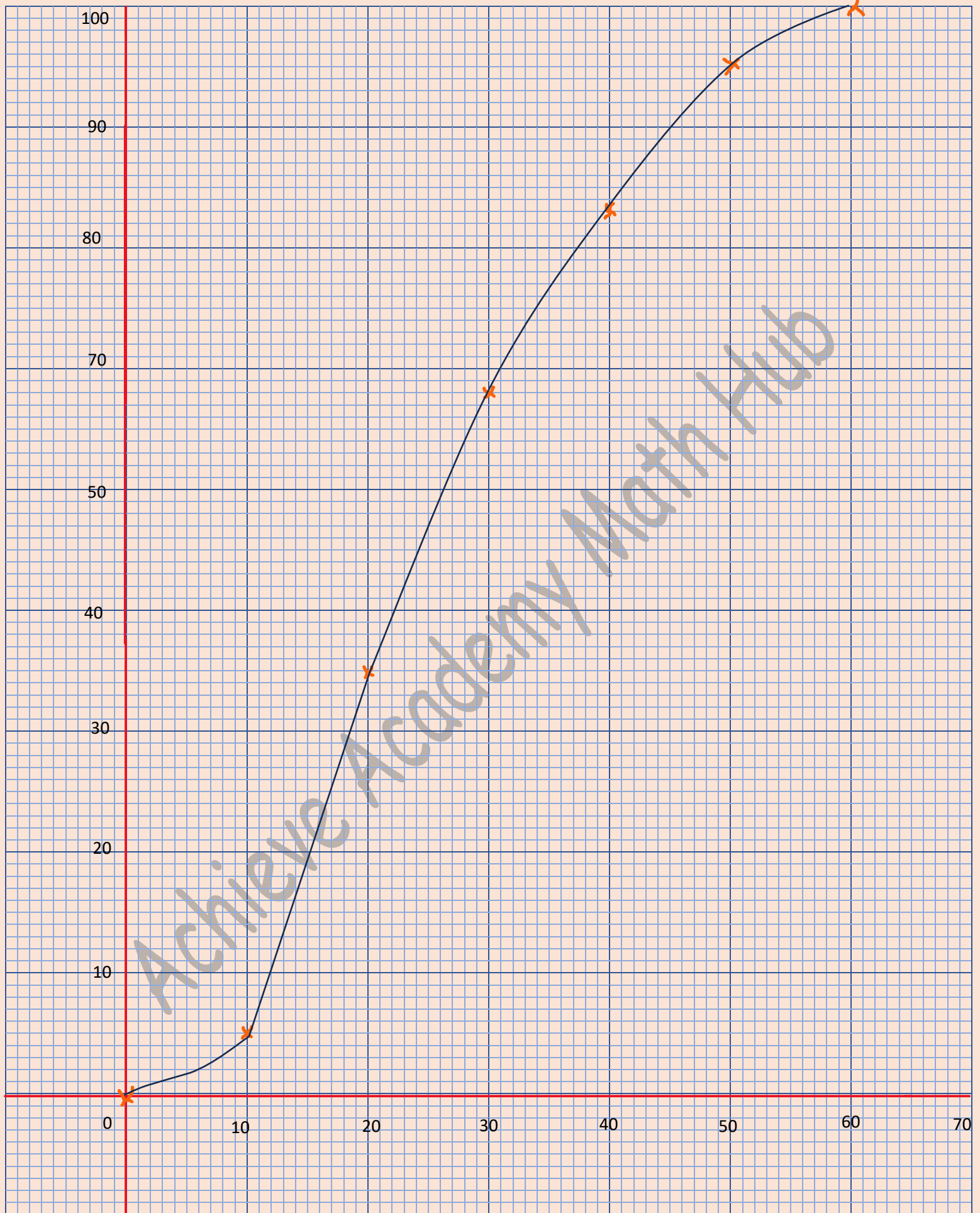
(b) Value of $q = \mathbf{95} \quad [1]$

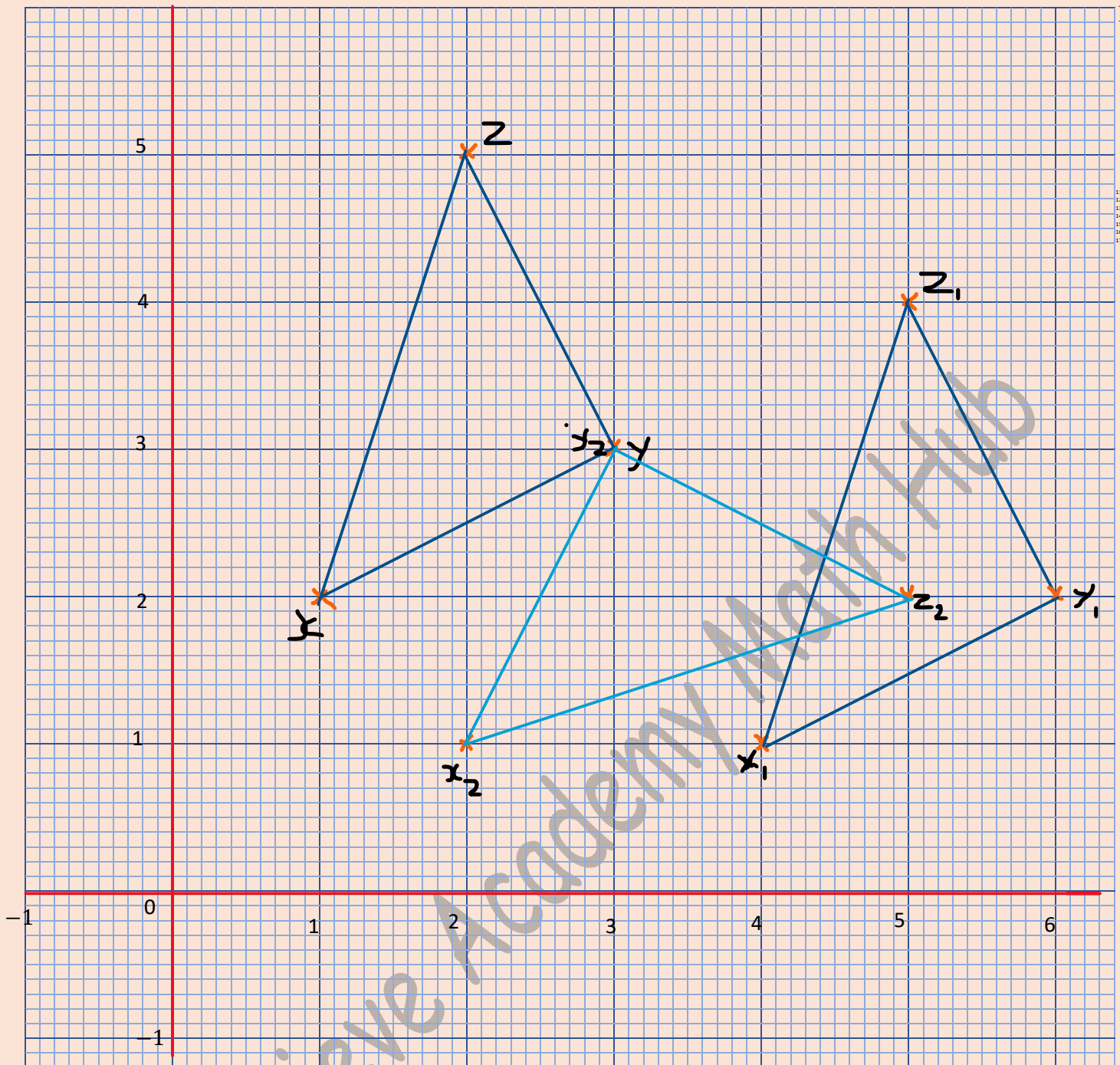
(d) (i) $\mathbf{73 \text{ customers}} \quad [2]$

(ii) Median Time = $\mathbf{26.5 \text{ minutes}} \quad [2]$

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(b) (i) Translation of vector $T = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$ [1]

(c) (ii) Description is *Reflection in the line $y = x$* [3]

(d) Draw Triangle $X_3Y_3Z_3$

$$X_3 = (-1; 2)$$

$$Y_3 = (-3; 3)$$

$$Z_3 = (-2; 5)$$

On the Graph [3]

12. (i) $\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$

$$= \vec{p} + \vec{q} \quad [1]$$

(ii) $\overrightarrow{OC} = 2\overrightarrow{OB}$

$$= 2(\vec{p} + \vec{q})$$

$$= 2\vec{p} + 2\vec{q} \quad [2]$$

(iii) $\overrightarrow{OM}$

$$\overrightarrow{AM} = \frac{1}{2}\overrightarrow{AB}$$

$$= \frac{1}{2}\vec{q}$$

$$\overrightarrow{OM} = \overrightarrow{OA} + \overrightarrow{AM}$$

$$= \vec{p} + \frac{1}{2}\vec{q} \quad [1]$$

(iv) $\overrightarrow{AC} = -\overrightarrow{OA} + \overrightarrow{OC}$

$$= -\vec{p} + 2\vec{p} + 2\vec{q}$$

$$= \vec{p} + 2\vec{q} \quad [1]$$

(b) $\overrightarrow{OT} = k\left(\vec{p} + \frac{1}{2}\vec{q}\right)$

$$= k\vec{p} + \frac{1}{2}k\vec{q} \quad [1]$$

(c) $\overrightarrow{OT} = \overrightarrow{OA} + \overrightarrow{AT}$

$$= \vec{p} + h(\vec{p} + 2\vec{q})$$

$$= (1 + h)\vec{p} + 2h\vec{q} \quad [1]$$

(d) value of h and k

$$\overrightarrow{OT} = \overrightarrow{OT}$$

$$k\vec{p} + \frac{1}{2}k\vec{q} = (1 + h)\vec{p} + 2h\vec{q}$$

$$k = 1 + h \quad (1)$$

$$\frac{1}{2}k = 2h \quad (2)$$

Using substitution method

$$\frac{1}{2}k = 2h$$

$$k = 4h$$

Now substitute k into the first equation

$$k = 1 + h$$

$$4h = 1 + h$$

$$\frac{3h}{3} = \frac{1}{3}$$

$$h = \frac{1}{3}$$

Now find k

$$k = 4 \left(\frac{1}{3} \right)$$

$$= \frac{4}{3}$$

$$\therefore h = \frac{1}{3} \text{ \& } k = \frac{4}{3} \quad [3]$$

(e) The ratio $MT:OT$

$$= \frac{1}{3} : \frac{4}{3}$$

$$= 1 : 4 \quad [2]$$

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OMISSIONS IS ALWAYS WELCOME

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But Jesus beheld them, & said unto them, "with men this is impossible; but
with GOD all things are possible" MATTHEW 19:26 (KJV)